

PROJECT (MYSQL)

BRANCHES

Table: branches

Column	Data Type	Constraints	Purpose
branch_id	INT	PK	Unique branch ID
branch_code	VARCHAR(10)	UNIQUE, NOT NULL	Bank branch code
branch_name	VARCHAR(100)	NOT NULL	Branch name
city	VARCHAR(50)	NOT NULL	City
state	VARCHAR(50)	NOT NULL	State
pincode	VARCHAR(10)		Pincode
address	VARCHAR(255)		Complete address
ifsc_code	VARCHAR(20)	UNIQUE, NOT NULL	Branch IFSC
phone	VARCHAR(15)		Branch phone
manager_name	VARCHAR(100)		Branch manager
opening_date	DATE		Branch opening date
status	VARCHAR(20)		ACTIVE/CLOSED

SQL

```
CREATE TABLE branches (  
    branch_id INT PRIMARY KEY,  
    branch_code VARCHAR(10) UNIQUE NOT NULL,  
    branch_name VARCHAR(100) NOT NULL,  
    city VARCHAR(50) NOT NULL,  
    state VARCHAR(50) NOT NULL,  
    pincode VARCHAR(10),  
    address VARCHAR(255),  
    ifsc_code VARCHAR(20) UNIQUE NOT NULL,  
    phone VARCHAR(15),  
    manager_name VARCHAR(100),  
    opening_date DATE,  
    status VARCHAR(20)  
);
```

CUSTOMERS

Table: customers

Column	Data Type	Constraints	Purpose
customer_id	INT	PK	Unique customer
customer_number	VARCHAR(20)	UNIQUE	Bank customer number
first_name	VARCHAR(50)	NOT NULL	First name
last_name	VARCHAR(50)	NOT NULL	Last name
email	VARCHAR(100)	UNIQUE	Email
phone	VARCHAR(15)	UNIQUE	Mobile number
date_of_birth	DATE		DOB
gender	VARCHAR(10)		Gender
address	VARCHAR(255)		Address
city	VARCHAR(50)		City
state	VARCHAR(50)		State
pincode	VARCHAR(10)		Pincode
customer_since	DATE		Registration date
customer_status	VARCHAR(20)		ACTIVE/INACTIVE/BLOCKED

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SQL

```
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY,  
    customer_number VARCHAR(20) UNIQUE NOT NULL,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    phone VARCHAR(15) UNIQUE,  
    date_of_birth DATE,  
    gender VARCHAR(10),  
    address VARCHAR(255),  
    city VARCHAR(50),  
    state VARCHAR(50),  
    pincode VARCHAR(10),  
    customer_since DATE,  
    customer_status VARCHAR(20)  
);
```

EMPLOYEES

Table: employees

Column	Data Type	Constraints	Purpose
employee_id	INT	PK	Employee ID
employee_code	VARCHAR(20)	UNIQUE	Employee code
branch_id	INT	FK	Employee branch
employee_name	VARCHAR(100)	NOT NULL	Employee name
job_role	VARCHAR(50)		Job designation
email	VARCHAR(100)	UNIQUE	Employee email
phone	VARCHAR(15)		Phone
salary	DECIMAL(12,2)		Salary
joining_date	DATE		Joining date
status	VARCHAR(20)		ACTIVE/INACTIVE

SQL

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    employee_code VARCHAR(20) UNIQUE NOT NULL,  
    branch_id INT,  
    employee_name VARCHAR(100) NOT NULL,  
    job_role VARCHAR(50),  
    email VARCHAR(100) UNIQUE,  
    phone VARCHAR(15),  
    salary DECIMAL(12,2),  
    joining_date DATE,  
    status VARCHAR(20),  
  
    FOREIGN KEY (branch_id)  
        REFERENCES branches(branch_id)  
);
```

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ACCOUNTS

A customer can have multiple accounts.

Table: accounts

Column	Data Type	Constraints	Purpose
account_id	INT	PK	Internal account ID
customer_id	INT	FK	Account owner
branch_id	INT	FK	Account branch
account_number	VARCHAR(20)	UNIQUE	Actual account number
account_type	VARCHAR(30)		SAVINGS/CURRENT/SALARY/FD
balance	DECIMAL(15,2)		Current balance
minimum_balance	DECIMAL(15,2)		Required minimum
opened_date	DATE		Account opening
status	VARCHAR(20)		ACTIVE/BLOCKED/CLOSED
currency	VARCHAR(10)		INR etc.

SQL

```
CREATE TABLE accounts (  
    account_id INT PRIMARY KEY,  
    customer_id INT NOT NULL,  
    branch_id INT NOT NULL,  
    account_number VARCHAR(20)  
        UNIQUE NOT NULL,  
    account_type VARCHAR(30)  
        NOT NULL,  
    balance DECIMAL(15,2)  
        DEFAULT 0,  
    minimum_balance DECIMAL(15,2)  
        DEFAULT 0,  
    opened_date DATE,  
    status VARCHAR(20),  
    currency VARCHAR(10)  
        DEFAULT 'INR',  
    FOREIGN KEY (customer_id)  
        REFERENCES customers(customer_id),  
    FOREIGN KEY (branch_id)  
        REFERENCES branches(branch_id)  
);
```

TRANSACTIONS

This should be your main analytics table.

It stores deposits, withdrawals, UPI transactions, ATM transactions, etc.

Table: transactions

Column	Data Type	Constraints	Purpose
transaction_id	BIGINT	PK	Unique transaction
account_id	INT	FK	Account involved
transaction_type	VARCHAR(30)		DEPOSIT/WITHDRAWAL/TRANSFER
transaction_mode	VARCHAR(30)		UPI/ATM/NEFT/IMPS/BRANCH
amount	DECIMAL(15,2)		Transaction amount
transaction_date	DATETIME		Date & time
description	VARCHAR(255)		Transaction description
reference_number	VARCHAR(50)	UNIQUE	Transaction reference

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balance_after	DECIMAL(15,2)		Balance after transaction
transaction_status	VARCHAR(20)		SUCCESS/FAILED/PENDING
location	VARCHAR(100)		Transaction location

SQL

```
CREATE TABLE transactions (  
    transaction_id BIGINT PRIMARY KEY,  
    account_id INT NOT NULL,  
    transaction_type VARCHAR(30) NOT NULL,  
    transaction_mode VARCHAR(30) NOT NULL,  
    amount DECIMAL(15,2) NOT NULL,  
    transaction_date DATETIME NOT NULL,  
    description VARCHAR(255),  
    reference_number VARCHAR(50)  
        UNIQUE,  
    balance_after DECIMAL(15,2),  
    transaction_status VARCHAR(20),  
    location VARCHAR(100),  
    FOREIGN KEY (account_id)  
        REFERENCES accounts(account_id)  
);
```

Example values
transaction_type:

DEPOSIT
WITHDRAWAL
TRANSFER
PAYMENT
REFUND
INTEREST
transaction_mode:

ATM
UPI
NEFT
RTGS
IMPS
BRANCH
NET_BANKING
MOBILE_APP

CARDS

Stores debit and credit cards.

Table: cards

Column	Data Type	Purpose
card_id	INT	Unique card
customer_id	INT	Card owner
account_id	INT	Linked account
card_number	VARCHAR(20)	Card number
card_type	VARCHAR(20)	DEBIT/CREDIT
card_network	VARCHAR(20)	VISA/MASTERCARD/RUPAY
issue_date	DATE	Issue date
expiry_date	DATE	Expiry
status	VARCHAR(20)	ACTIVE/BLOCKED/EXPIRED

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credit_limit	DECIMAL(15,2)	Credit limit
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SQL

```
CREATE TABLE cards (  
    card_id INT PRIMARY KEY,  
    customer_id INT NOT NULL,  
    account_id INT,  
    card_number VARCHAR(20)  
        UNIQUE NOT NULL,  
    card_type VARCHAR(20),  
    card_network VARCHAR(20),  
    issue_date DATE,  
    expiry_date DATE,  
    status VARCHAR(20),  
    credit_limit DECIMAL(15,2),  
    FOREIGN KEY (customer_id)  
        REFERENCES customers(customer_id),  
    FOREIGN KEY (account_id)  
        REFERENCES accounts(account_id)  
);
```

CARD TRANSACTIONS

This table stores purchases made through cards.

Table: card_transactions

Column	Data Type	Purpose
card_transaction_id	BIGINT	Unique transaction
card_id	INT	Card used
amount	DECIMAL(15,2)	Purchase amount
merchant_name	VARCHAR(100)	Merchant
merchant_category	VARCHAR(50)	Food/Shopping etc.
transaction_date	DATETIME	Date/time
city	VARCHAR(50)	Transaction city
transaction_status	VARCHAR(20)	SUCCESS/FAILED
reference_number	VARCHAR(50)	Reference
payment_channel	VARCHAR(30)	POS/ONLINE/CONTACTLESS

SQL

```
CREATE TABLE card_transactions (  
    card_transaction_id BIGINT PRIMARY KEY,  
  
    card_id INT NOT NULL,  
  
    amount DECIMAL(15,2) NOT NULL,  
  
    merchant_name VARCHAR(100),  
  
    merchant_category VARCHAR(50),  
  
    transaction_date DATETIME,  
  
    city VARCHAR(50),
```

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transaction_status VARCHAR(20),

reference_number VARCHAR(50),

payment_channel VARCHAR(30),

FOREIGN KEY (card_id)

REFERENCES cards(card_id)

);

BENEFICIARIES

When a customer adds another bank account for transferring money, it becomes a beneficiary.

Table: beneficiaries

Column	Data Type	Purpose
beneficiary_id	INT	Unique beneficiary
customer_id	INT	Customer who added beneficiary
beneficiary_name	VARCHAR(100)	Beneficiary name
account_number	VARCHAR(30)	Beneficiary account
bank_name	VARCHAR(100)	Bank
ifsc_code	VARCHAR(20)	IFSC
nickname	VARCHAR(50)	Personal label
added_date	DATE	Added date
status	VARCHAR(20)	ACTIVE/BLOCKED

SQL

CREATE TABLE beneficiaries (

beneficiary_id INT PRIMARY KEY,

customer_id INT NOT NULL,

beneficiary_name VARCHAR(100) NOT NULL,

account_number VARCHAR(30) NOT NULL,

bank_name VARCHAR(100),

ifsc_code VARCHAR(20),

nickname VARCHAR(50),

added_date DATE,

status VARCHAR(20),

FOREIGN KEY (customer_id)

REFERENCES customers(customer_id)

);

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TRANSFERS

This table represents money moving between two accounts.

Table: transfers

Column	Data Type	Purpose
transfer_id	BIGINT	Unique transfer
sender_account_id	INT	Sender
receiver_account_id	INT	Receiver
amount	DECIMAL(15,2)	Transfer amount
transfer_mode	VARCHAR(30)	UPI/IMPS/NEFT/RTGS
transfer_date	DATETIME	Date/time
reference_number	VARCHAR(50)	Transaction reference
transfer_status	VARCHAR(20)	SUCCESS/FAILED/PENDING
remarks	VARCHAR(255)	Description

SQL

```
CREATE TABLE transfers (  
    transfer_id BIGINT PRIMARY KEY,  
  
    sender_account_id INT NOT NULL,  
  
    receiver_account_id INT NOT NULL,  
  
    amount DECIMAL(15,2) NOT NULL,  
  
    transfer_mode VARCHAR(30),  
  
    transfer_date DATETIME,  
  
    reference_number VARCHAR(50)  
        UNIQUE,  
  
    transfer_status VARCHAR(20),  
  
    remarks VARCHAR(255),  
  
    FOREIGN KEY (sender_account_id)  
        REFERENCES accounts(account_id),  
  
    FOREIGN KEY (receiver_account_id)  
        REFERENCES accounts(account_id)  
);
```